# EXPERIMENT 2: SIMULATION OF NFA

## Aim

To write a C program to simulate a Non-Deterministic Finite Automaton (NFA) that accepts binary strings starting with `0` and ending with `1`.

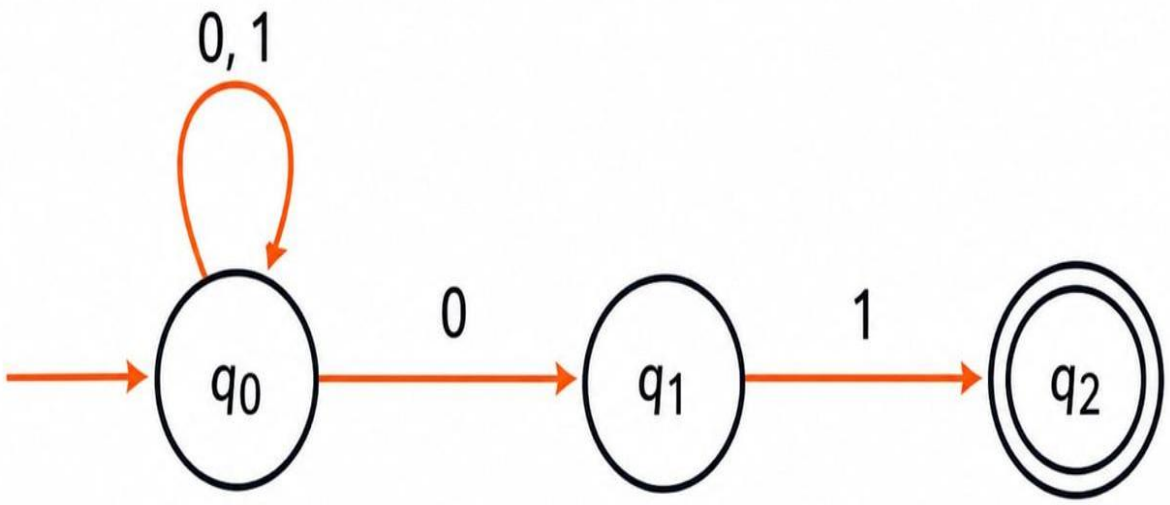
## Procedure / Algorithm

1. Start the program.
2. Read a binary string from the user.
3. Check whether the first character is `0`.
4. Check whether all characters are either `0` or `1`.
5. Check whether the last character is `1`.
6. If all conditions are satisfied, print "String Accepted"; otherwise, print "String Rejected".
7. Stop the program.

## Example:

A Non-Deterministic Finite Automaton (NFA) accepts a binary string if it starts with 0 and ends with 1. The input alphabet is Sigma={0,1}. An NFA can have multiple possible transitions for the same input symbol from one state.

## Sample Design:



## Transition Table:

| Present state | Input 0 | Input 1 |
|---|---|---|
| → q_0 | q_1 | nil |
| q_1 | q_1 | q_1,q_2 |
| *q_2 | nil | nil |

## Code:

```c
#include <stdio.h>
#include <string.h>
int main()
{
    char s[100];
    int i, n, q0 = 1, q1 = 0, q2 = 0;
    int a, b, c;
    printf("Enter a binary string: ");
    scanf("%99s", s);
    n = strlen(s);
    for (i = 0; i < n; i++)
    {
        a = b = c = 0;
        if (q0 && s[i] == '0')
            b = 1;
        if (q1 && (s[i] == '0' || s[i] == '1'))
            b = 1;
        if (q1 && s[i] == '1')
            c = 1;
        q0 = a;
        q1 = b;
        q2 = c;
    }
    if (q2)
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

```
Enter a binary string: 001
String Accepted
```

## Input / Output

Input1: Text Enter a binary string: 001 Output1:

Text:String Accepted Input2:Enter a binary string: 010

Output2:String Rejected

## Result

Thus, the C program to check binary strings starting with `0` and ending with `1` was executed successfully.